Mapping Irrigated Area at Field Scale Based on the OPtical TRApezoid Model (OPTRAM) Using Landsat Images and Google Earth Engine

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***Abstract*— Irrigation is critical to agricultural production in arid and semiarid regions, and it is imperative to map high-resolution irrigated area to improve water productivity. This study proposes a field-scale (30-m resolution) irrigated area mapping method based on soil moisture change detection using remote sensing data only. First, normalized soil moisture is obtained using the optical trapezoid model (OPTRAM) and then converted to soil water content. Next, individual irrigation events are identified in the time series of soil water content using threshold detection. Finally, irrigation events are accumulated over the time series, and then, the irrigated area map can be obtained. This method was tested using Google Earth Engine (GEE) to analyze remote sensing images and map irrigated areas in a typical arid and semiarid region called Hexi Corridor in northwestern China in the past 30 years. *In situ* validation shows that this method has an accuracy close to 100%. The shortcoming of low recall is also overcome by long-term observations. An application of the proposed method shows that the irrigated cropland of Hexi Corridor has increased by 4840 km^2 (42.2%) over a 31-year time period (1990–2020). This field-scale irrigated area mapping method can improve the management of water resources.**

***Index Terms*— Irrigation, remote sensing, soil moisture, time series.**

I. Introduction

IRRIGATED agriculture is of great significance to human societies, especially in arid and semiarid areas where irrigation provides the moisture required for the growth of crops [1]. Global land cover and land use have rapidly changed as a consequence of economic and social development [2]. In the 20th century, irrigated agricultural areas, which accounted for 18% of the total cropland area, quadrupled [3], contributed 40% of the total global food production, and consumed 70% of the total fresh water withdrawals [4], [5]. Hence, monitoring the extent and change of irrigated area is very important to food security and water resources management.

Remote sensing is considered to be an effective tool for mapping irrigated area [6]. Based on remote sensing, several global irrigated area mapping products with low resolution (10–25 km) are available, such as the global irrigated area map (GIAM) [7] and the Historical Irrigation dataset (HID) [3]. However, these low spatial resolution datasets provided estimates of irrigated area at a global scale but are not suitable for applications at field scale (∼30 m) [8]. For the coarse resolution, data tend to include more mixed pixels, often resulting in low accuracy at the edge of the fields [9] and excluding small settlements and villages embedded in cropland areas. In view of this, it is necessary to develop irrigated area maps at field scale to improve the management of irrigation.

There are several irrigated area datasets with higher spatial resolution (30–250 m) than these coarse-resolution datasets at the regional scale. Such as, using Landsat and Moderate Resolution Imaging Spectroradiometer (MODIS), the Landsat-based Irrigation Dataset (LANID) [10], MODIS-based Irrigated Agriculture Dataset for the U.S. (MIrAD-US) [11], and a combined product based on a two-step mapping method that uses both MODIS and Landsat data [12] provide more details on the spatial distribution of the irrigated area. These datasets are based on supervised classifications (e.g., random forest [10] and support vector machine [12]) and data assimilation of statistics and remote sensing data [11]. In these models, the commonly used variables are surface reflectance, land surface temperature (LST), vegetation index (VI), digital elevation model (DEM), and precipitation. These methods heavily rely on accurate training data or statistics data to establish the relationship between variables and irrigation status, but these data may be difficult to obtain, especially in developing countries. Therefore, a method entirely based on remote sensing is needed to map global irrigated area at field scale.

In addition to the above methods, irrigation's physical effects can also be used to map irrigated area. Irrigation provides better crop conditions than nonirrigation in growing seasons [13]. The irrigation cooling effect on LST is a well-documented phenomenon [14] and was used to map irrigated

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area in a recent study [15]. However, this method still cannot be used for mapping irrigated area at field scale, limited by the resolution of thermal data (e.g., 100 m for Landsat-8 and 1000 m for MODIS). In addition to the reduction of LST, irrigation for cropland can lead to a significant increase in soil moisture [16]. Recent studies have shown that satellite-based soil moisture with coarse resolution can be used to identify irrigation [17], [18]. Threshold detection provides a simple but powerful way to identify land surface changes [19]. In view of this, soil moisture datasets could be used to map irrigated area.

The use of passive microwave data is currently the most accurate way for soil moisture retrieval [20], [21], and its coarse spatial resolution can be improved to 1 km by several downscaling methods [22]. These methods are: 1) machine learning methods to establish a relationship between remote sensing data (e.g. MODIS) and surface soil moisture (SSM) datasets [23], in which training data are SSM with coarse resolution, and 2) and land surface model for hydrological simulation [24]. The optical trapezoid model (OPTRAM) provides another approach for obtaining high-resolution soil moisture [25] and shows potential in agricultural applications [26]. It was evaluated that OPTRAM performs better than the advanced scatterometer (ASCAT) and provided the mean unbiased root-mean-squared error (ubRMSE) of 0.0675 $cm^3 \cdot cm^{-3}$ for different underlying surfaces [27]. Therefore, OPTRAM could be applied to directly map irrigated areas from the increasement of soil moisture at field scale [28], [29].

Combining OPTRAM with change detection, this study proposes an irrigated area mapping method at field scale. This method consists of three steps. First, soil water content is obtained using the OPTRAM model. Next, individual irrigation events are identified using the threshold detection of soil water content. Finally, the irrigated area is mapped as the accumulation of irrigation events. The novelty of this study is that this method is: 1) completely based on remote sensing data and does not rely on any statistics or manual interpretation and 2) based on physical effects of irrigation instead of spectral analysis. The detailed map of irrigated area will provide better data quality and more accurate analyzing results for water resources management and food security.

II. Materials

A. Study Area

The study area is the cropland of Hexi Corridor, which consists of three basins: Heihe River Basin, Shiyang River Basin, and Shule River Basin (Fig. 1). Located in the vast and sparsely populated northwestern China, Hexi Corridor is one of the main human settlements. Its artificial oases along the narrow alluvial plain produce wheat, barley, maize, and other crops. Hexi Corridor covers an area of 214 616 km^2, and its elevation ranges from 788 m to 5821 m above sea level. With an average annual precipitation of about 150 mm, the study area is a typical arid and semiarid region. Thus, its agriculture counts heavily on irrigation from groundwater and river water sources. Hexi Corridor is one of the first areas in China to develop large-scale facilities for irrigated agriculture and has become a representative case of oasis agriculture in northwestern China. Under the profound influence of the natural environment and human activities, Hexi Corridor has been widely used to study human–environment interactions and surface processes in arid and semiarid regions. For more information about the study area, readers are kindly referred to [30]–[33].

B. In Situ Data

SoilNET was an ecohydrological wireless sensor network in the midstream of the Heihe River Basin [34]. It consisted of 51 soil moisture observation nodes based on the principle of time-domain transmissometry (TDT) and was densely distributed in an area of 5.5 km × 5.5 km mainly covered by maize. Soil moisture and soil temperature at depths of 4, 10, 20, and 40 cm were observed every 10 min at each node from June to August 2012. The data were mainly used to capture and analyze soil moisture's temporal and spatial variability at a small scale (∼30 m) [35]–[37].

A meteorological observation system named Daman superstation in the Heihe River Basin provided long-term *in situ* measurements of soil moisture [38]. Soil moisture at depths of 2, 4, 10, 20, and 40 cm was recorded every 10 min from 2013 to 2019. This system can capture long-term changes in soil moisture [39]. Another four nodes with TDT were installed in the Shiyang River Basin. Soil moisture at depths of 20, 40, 60, 80, and 100 cm was measured every 30 min from 2015 to 2018.

Because of the lack of first-hand records for irrigation events, ground-truth irrigation events were extracted from *in situ* soil moisture measurements for validation. *In situ* measurements for soil moisture (Table I) were resampled at a daily scale by taking the closest record to 11:30 am, which is also the local time of Landsat satellite overpass. Daily soil water content was obtained by multiplying the soil moisture by the soil thickness per observing layer as follows:

$$W = \sum_{i=1}^{n} T_i \theta_i \tag{1}$$

where W is soil water content (mm), T_i is the soil thickness (mm) of layer i, and θ_i is the soil moisture ($cm^3 \cdot cm^{-3}$) of layer i. Finally, threshold detection was used to obtain ground-truth of irrigation events as

$$D = \Delta W - P \tag{2}$$

where ΔW is the daily variation of soil water content (mm), P is the precipitation (mm), and D is the unnatural increase of *in situ* soil water content. If $D > 10$ mm, it is considered that an irrigation event has occurred. Then, the date was recorded as the ground-truth irrigation date.

C. Satellite Data

The Landsat mission has been collecting high-resolution data of the land surface since 1972 and is regarded as the longest continuous mission of its kind. The Operational Land Imager (OLI) on the Landsat-8, the Enhanced Thematic

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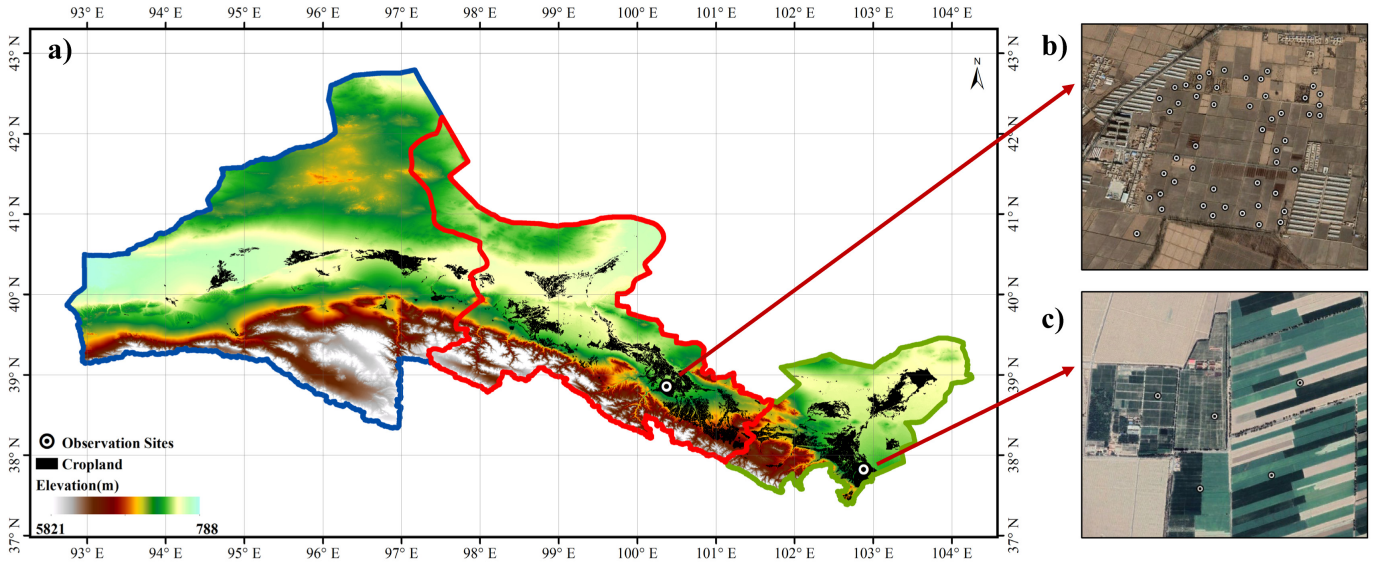


Fig. 1. Elevation from the Shuttle Radar Topography Mission (SRTM) and distribution of cropland in Hexi Corridor, which consists of three basins: Shule River Basin (blue), Heihe River Basin (red), and Shiyang River Basin (green).

Mapper Plus (ETM+) on the Landsat-7, and the Thematic Mapper (TM) on the Landsat-5 can provide global land surface reflectance with a high spatial resolution of 30 m and a temporal resolution of 16 days, and it is widely used in geoscience research. The Soil Moisture Active Passive (SMAP) mission has been measuring SSM since 2015. SMAP provides global SSM products with an accuracy better than 0.04 cm^3/cm^3 [40]. Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) is a quasi-global precipitation dataset with a spatial resolution of 0.05° available from 1981 to the present [41]. CHIRPS combines satellite images with site observations to provide gridded data of daily precipitation that can be used for global precipitation trend analysis and seasonal drought monitoring. Information of these datasets are listed in Table I.

This study used the SMAP L3 global SSM product and land surface reflectance from Landsat-5, Landsat-7, and Landsat-8 to fit key parameters of the soil moisture retrieval method used in this article. The quality band of Landsat was used to eliminate the pixels contaminated by clouds, cloud shadows, ice/snow, and water bodies. To avoid false irrigation detections caused by precipitation, the CHIRPS precipitation product was used to evaluate the influence of rainfall on soil water content. Nearest neighbor interpolation is used to obtain 30-m precipitation because precipitation generally has low spatial heterogeneity. All of the satellite data used for the study were obtained via Google Earth Engine (GEE), a free planetary-scale cloud computing platform designed for geospatial science [42]. The authors developed several programs to map satellite-based irrigated area and all of these programs were coded in JavaScript with GEE Code Editor.

D. Other Materials

GlobeLand30 (http://www.globallandcover.com/) is a global geoinformation public product for land cover with 30-m resolution in 2000, 2010, and 2020, mapping water body, wetland, artificial region, tundra, ice, grassland, bare land, cropland, shrub, and forest [43]. This study used GlobeLand30 in 2000, 2010, and 2020 to map the maximum extent of cropland in Hexi Corridor, providing a base map of irrigated cropland.

III. Methods

To map the irrigated area at field scale, this study proposed a method based on soil moisture change detection using the OPTRAM model with remote sensing data only (Fig. 2), including three steps: soil water content retrieval (Section III-A), irrigation events detection (Section III-B), and irrigated area mapping (Section III-C). Validation strategies used in this study will be explained in Section III-D.

A. Retrieving Soil Water Content

OPTRAM, which was proposed by Sadeghi *et al.* [25] and Babaeian *et al.* [27], is a physically based soil moisture retrieval method using only the red, near-infrared (NIR), and shortwave-infrared (SWIR) bands, which are common to many satellite sensors. In the OPTRAM model, surface soil water content (W, top 5 cm) can be determined as

$$W = W_d + \theta(W_w - W_d) \tag{3}$$

where W_w and W_d represent the maximum and minimum surface soil water content, respectively; θ is called the normalized soil moisture and is given as

$$\theta = \frac{\mathrm{STR} - \mathrm{STR}_d}{\mathrm{STR}_w - \mathrm{STR}_d} \tag{4}$$

where STR is the SWIR transformed reflectance and is related to land surface reflectance in the SWIR band, which is given as

$$\mathrm{STR} = \frac{(1 - R_{\mathrm{SWIR}})^2}{2R_{\mathrm{SWIR}}} \tag{5}$$

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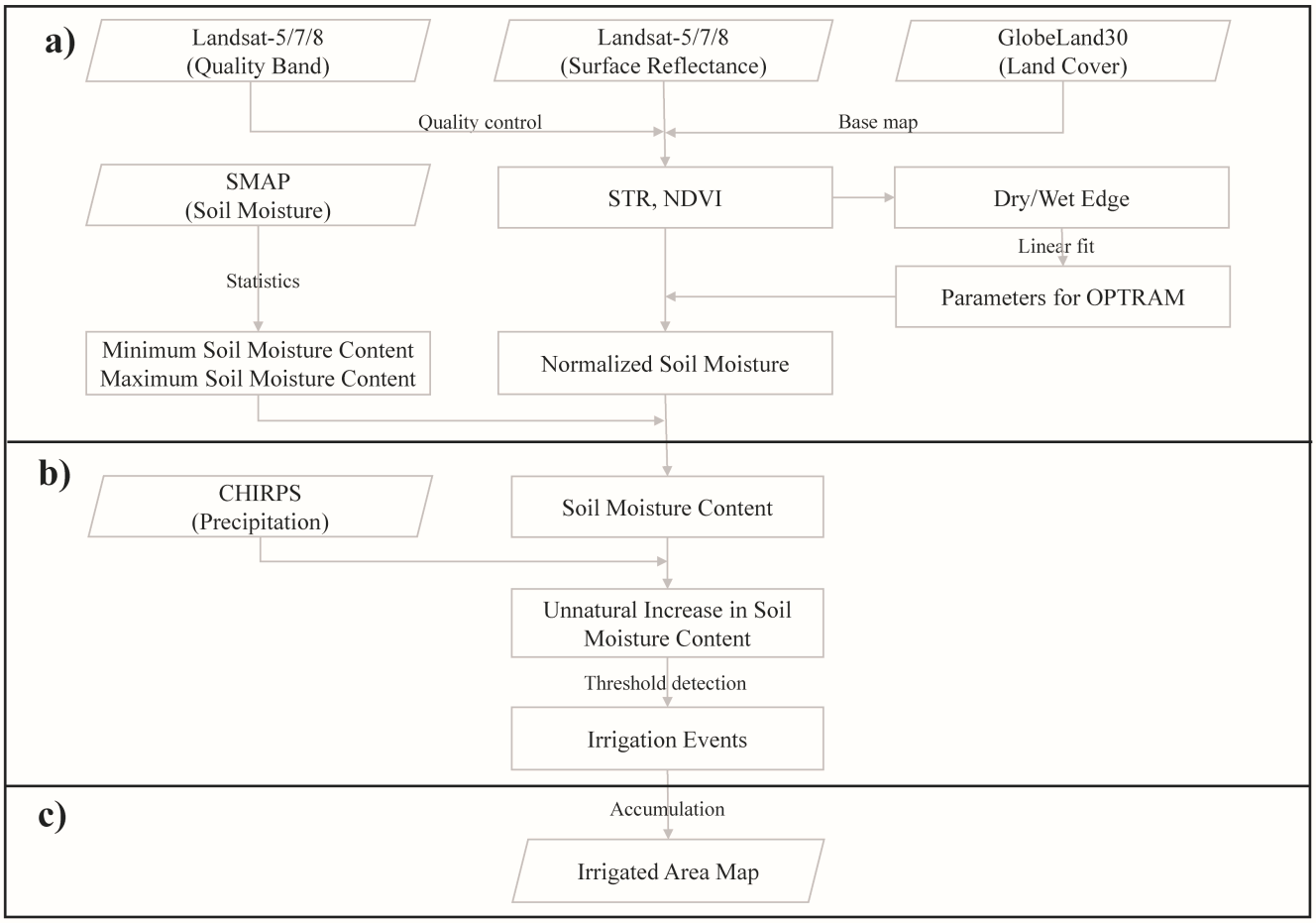


Fig. 2. Irrigated area mapping method: (a) soil water content retrieval based on the OPTRAM, (b) threshold-based detection of irrigation events, and (c) irrigated area mapping.

TABLE I
LIST OF DATA USED IN THIS STUDY

Name	Type	Spatial Resolution	Temporal Resolution	Time Available
Landsat-5 TM	Land surface reflectance	30 m	16-day	1990-1999
Landsat-7 ETM+	Land surface reflectance	30 m	16-day	2012.6-2012.8
Landsat-8 OLI	Land surface reflectance	30 m	16-day	2014-2020
SMAP	Satellite-based SM	10 km	3-day	2015-2020
CHIRPS	Precipitation	0.05°	1-day	1990-2020
GlobeLand30	Land cover	30 m	--	2000, 2010, 2020
Shiyang	In situ SM	--	30-minute	2015-2018
SoilNET	In situ SM	--	10-minute	2012.6-2012.8
Daman	In situ SM	--	10-minute	2013-2019

and STR_w and STR_d represent the maximum and minimum STR in the specific vegetation condition, respectively, as shown in Fig. 3. The most common VI used in OPTRAM is the normalized difference vegetation index (NDVI) that is given as

$$\mathrm{NDVI} = \frac{R_{\mathrm{NIR}} - R_{\mathrm{red}}}{R_{\mathrm{NIR}} + R_{\mathrm{red}}} \tag{6}$$

where R_{NIR} and R_{red} are the land surface reflectance in the NIR and red bands, respectively. Both STR and NDVI are calculated from the Landsat series. Then, STR_w and STR_d are calculated as

$$\mathrm{STR}_d = i_d + s_d \mathrm{NDVI} \tag{7}$$

$$\mathrm{STR}_w = i_w + s_w \mathrm{NDVI} \tag{8}$$

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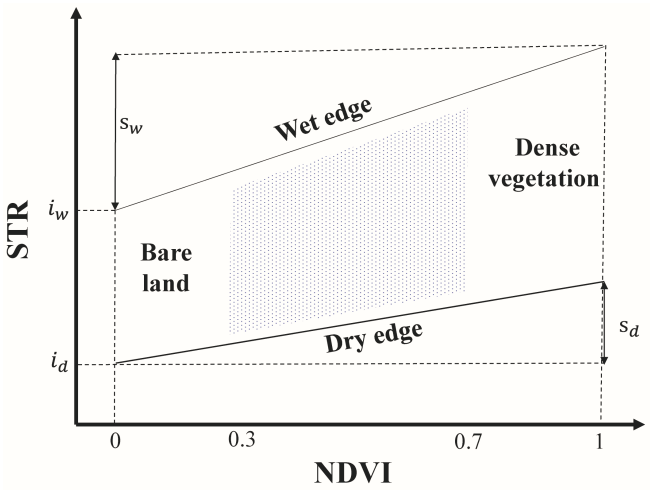


Fig. 3. Theory of OPTRAM.

where i is the intercept on the STR axis, s is the slope, the subscript "d" represents the dry edge, and the subscript "w" represents the wet edge.

Combining the equations above, the surface soil water content for each pixel can be estimated as a function of STR and NDVI

$$W = W_d + \frac{(i_d + s_d\text{NDVI} - \text{STR})(W_w - W_d)}{i_d - i_w + (s_d - s_w)\text{NDVI}}. \tag{9}$$

In this study, Hexi Corridor was divided into 25 939 small areas of $0.01^\circ \times 0.01^\circ$, and the calculation of each area was performed independently. In each small area, six parameters are needed to predict soil water content: W_w, W_d, i_d, s_d, i_w, and s_w.

The maximum and the minimum soil water content of SMAP from 2015 to 2020 are regarded as W_w and W_d, respectively, which constrain the range of soil water content. This range in a small area ($0.01^\circ \times 0.01^\circ$) is thought to be consistent.

The NDVI axis was divided into 40 intervals from 0.3 to 0.7. For each small area, 80 pixels from 40 NDVI intervals were selected from more than 1 000 000 pixels in two steps: 1) for each NDVI interval and each image, pixels with 95th- and 5th-percentile values of STR are retained, and 2) for each NDVI interval, pixels with the largest and smallest value of STR are retained. The selected 80 pixels constituted the dry edge and wet edge that were shown in Fig. 3. Parameters, i_d, s_d, i_w, and s_w, were fitted by unary linear regression. Landsat images within two years are used for the above-mentioned parameter fitting for different satellites: 1994–1995 for Landsat-5, 2011–2012 for Landsat-7, and 2015–2016 for Landsat-8.

B. Detecting Irrigation Events

Irrigation and precipitation constitute the main water supply for crop growth in arid and semiarid areas. When ignoring the impact of runoff due to insufficient precipitation, the water balance equation in the arid and semiarid areas can be expressed as

$$\text{Ir} + P = \text{ET} + \Delta W + \text{In} \tag{10}$$

where P is the precipitation (mm), Ir is the irrigated water (mm), ET is the water loss via evapotranspiration (mm), ΔW is the variation of surface soil water content (mm), and In is the infiltration (mm). It is still difficult to accurately estimate high-resolution ET and In in the irrigated area [44], especially without 30-m resolution thermal infrared data. In semiarid areas, the instantaneous drastic changes in soil moisture caused by irrigation will gradually disappear within a few days. Considering that the goal of this study is to map the irrigated area rather than accurately retrieve the volume of irrigation, neglecting ET and In can make the method stronger and reduce the computational cost. Hence, (10) could be rewritten as

$$\text{Ir} > \Delta W - P. \tag{11}$$

Equation (11) shows that irrigation is larger than the difference between the variation of surface soil water content and precipitation. If the increased soil water content is greater than precipitation, such an unexpected increase in soil water content can only be caused by irrigation.

In this study, threshold detection was used to extract satellite-based irrigation events from the time series of surface soil water content

$$D = \Delta W - P \tag{12}$$

where D is the unnatural increase of satellite-based surface soil water content. The total precipitation within 16 days in a small area ($0.01^\circ \times 0.01^\circ$) is thought to be consistent. Considering that the accuracy of satellite-based retrieval of soil moisture is better than 0.1 $\text{cm}^3 \cdot \text{cm}^{-3}$ [27] and the thickness of the surface soil is 5 cm, we set an empirical threshold of 5 mm for D. If $D > 5$ mm for two temporally sequential images, it is considered that an irrigation event has occurred in this period. This period in which soil water content increases rapidly is recorded and a satellite-based irrigation event is counted.

C. Mapping Irrigated Area

Ground-truth irrigation events and satellite overpasses are mutually independent. Over a long period, many irrigation events will occur (about 5–6 times per year in regularly irrigated area of Hexi Corridor). This provides a potential way to map the entire irrigated area using long-term satellite observations. The first key issue is knowing how long observations are enough to map it.

Assume that p is the probability of an irrigated event, which can be detected by the proposed method

$$p = \frac{n}{N} \tag{13}$$

where n is the number of detected irrigation events using the proposed method and N is the number of total irrigation events. Hence, the probability that an irrigated pixel is omitted is given as

$$P_m = (1 - p)^N \tag{14}$$

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where P_m is the probability that an irrigated pixel is omitted. Assuming that $p = 0.1$ and $P_m < 0.01$, then $N > 43.7$, that is, if the probability that irrigation events detected by satellite observation is only 0.1 and if the area is irrigated for more than 44 times, there is only a probability of 0.01 that the irrigated pixel is omitted. In Hexi Corridor, irrigation times within 7–10 years can meet this requirement for the number of irrigation events. For an irrigated pixel, the expectation of satellite-based irrigation events (E_s) is

$$E_s = pN_{\text{gt}} \tag{15}$$

where N_{gt} is the number of ground-truth irrigation events within this pixel, that is, pixels where more irrigation events are detected are more likely to be irrigated frequently.

Another key issue is to know the possibility of false detection. Assume that the number of irrigation events (X) detected in a pixel follows a Poisson distribution. For a nonirrigated pixel, the probability of false detection is given as

$$P_f = 1 - \frac{\lambda^0}{0!}e^{-\lambda} \tag{16}$$

where P_f is the probability of false detection and λ is the probability that a satellite-based irrigation event does not match any true irrigation event. Assuming that $P_f < 0.01$, then $\lambda < 0.01$.

In this study, the irrigated area map is obtained by counting the detected irrigation events over a long period (1990–1999 for Landsat-5 and 2014–2020 for Landsat-8) to offset the missed detections caused by ignoring ET and In.

D. Validation Strategies

The accuracy of the satellite-based irrigation events detection was evaluated in the spatial and temporal dimensions because it greatly affects this irrigated area mapping method.

The date was recorded as the ground-truth irrigation data for *in situ* data. For satellite-based data, the dates of two temporally sequential images constituted a satellite-based irrigation event. If there was at least one ground-truth irrigation event in the period of a satellite-based irrigation event, the satellite-based irrigation event matched the ground truth.

An irrigated pixel is a positive sample when an irrigation event occurred and a negative sample when there was no irrigation event. If the detected irrigation events are accurate, the irrigated area mapping method approached in this article was deemed reliable.

In this study, precision and recall rate are used to evaluate the accuracy of detected irrigation events and irrigated pixels

$$P_p = \frac{\text{TP}_p}{\text{TP}_p + \text{FP}_p} \tag{17}$$

$$P_e = \frac{\text{TP}_e}{\text{TP}_e + \text{FP}_e} \tag{18}$$

$$R_p = \frac{\text{TP}_p}{\text{TP}_p + \text{FN}_p} \tag{19}$$

$$R_e = \frac{\text{TP}_e}{\text{TP}_e + \text{FN}_e} \tag{20}$$

where P is the precision, R is the recall rate, subscript "p" represents irrigated pixels, subscript "e" represents irrigation events, TP is the size of the intersection of the satellite-based irrigation and ground-truth irrigation, FN is the size of the difference set of the ground-truth irrigation and satellite-based irrigation, and FP is the size of the difference set of the satellite-based irrigation and ground-truth irrigation, that is, TP + FN is the size of the ground-truth irrigation and TP + FP is the size of the satellite-based irrigation.

TABLE II
COMPARISON BETWEEN SATELLITE-BASED AND *in Situ* IRRIGATION EVENTS FOR SOILNET

Item	Irrigated Station	Irrigated Events (station·time)
In situ	51	241
Satellite	34	34
Matched	34	34

IV. RESULTS

A. Validation

In the spatial dimension, *in situ* SoilNET provided a short-term SSM observation from June to August 2012. From June 24 to August 30, the cropland where SoilNET was located was irrigated at different times. Landsat-7 overpassed the region on June 24 and July 10. Table II shows that satellite-based irrigation events were detected in 34 stations, and all of them matched ground-truth irrigation events, though 33.3% of the irrigated stations and 85.9% of the irrigation events were missed. In this short period, both P_p and P_e are 100%, R_p is 66.7%, and R_e is 14.1%.

In the temporal dimension, the Daman superstation (2013–2019) in the Heihe River Basin and four stations with TDT (2015–2018) in the Shiyang River Basin provided long-term SSM measurements. We extracted 94 ground-truth irrigation events from *in situ* soil moisture data using the method described in Section II-B and 13 satellite-based irrigation events from Landsat-8 data using the proposed method. All of the satellite-based events are correctly detected. Fig. 4 shows the result of Daman superstation, in which four satellite-based events match the ground-truth events. Thus, in this long period, P_p, P_e, and R_p are 100%, and R_e is 13.8%.

In addition, both satellite-based and ground-truth events show that there were more irrigation events in summer. In the arid and semiarid areas, irrigation contributes more to the growth of crops than precipitation. Many irrigation events are expected to be omitted due to the strict selection criteria, but areas where irrigation occurs frequently are more likely to be detected over a long period. It is worth mentioning that cloud occlusion has a great influence on optical remote sensing. On these five stations, continuous images without cloud only accounted for 49.9%, that is, about half of ground-truth irrigation events are undetectable for satellite observation.

The validation in spatial and temporal dimensions shows that the proposed method has similar performance and the accuracy is very high (here is 100%).

B. Spatial Distribution

Although many irrigation events during this period were missed, the method proposed in this article was highly accurate

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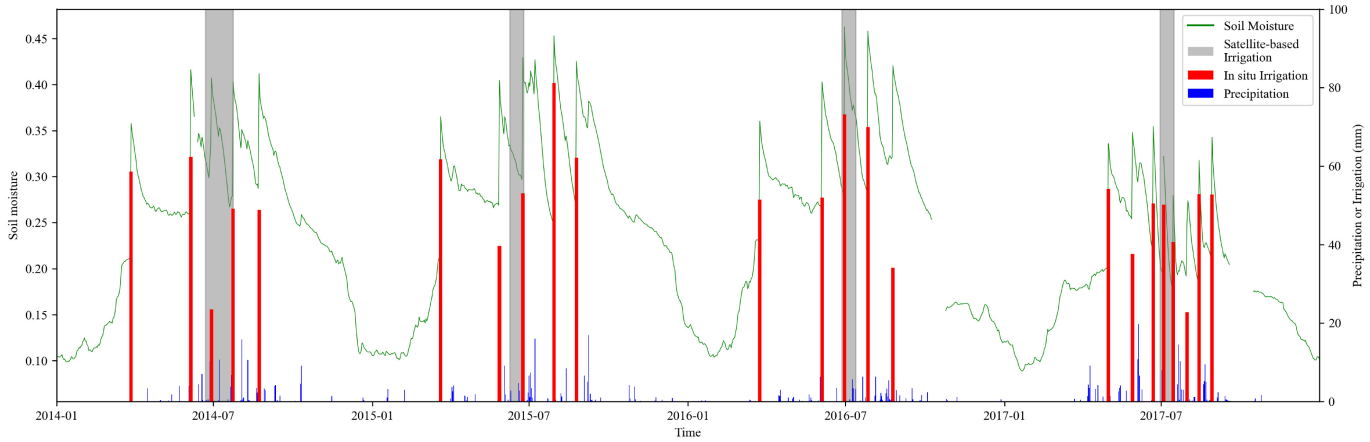


Fig. 4. Soil moisture measurements in Daman superstation during 2014–2017. Green plot: SSM. Red bar: ground-truth irrigation events. Blue bar: precipitation. Gray rectangles: satellite-based irrigation events.

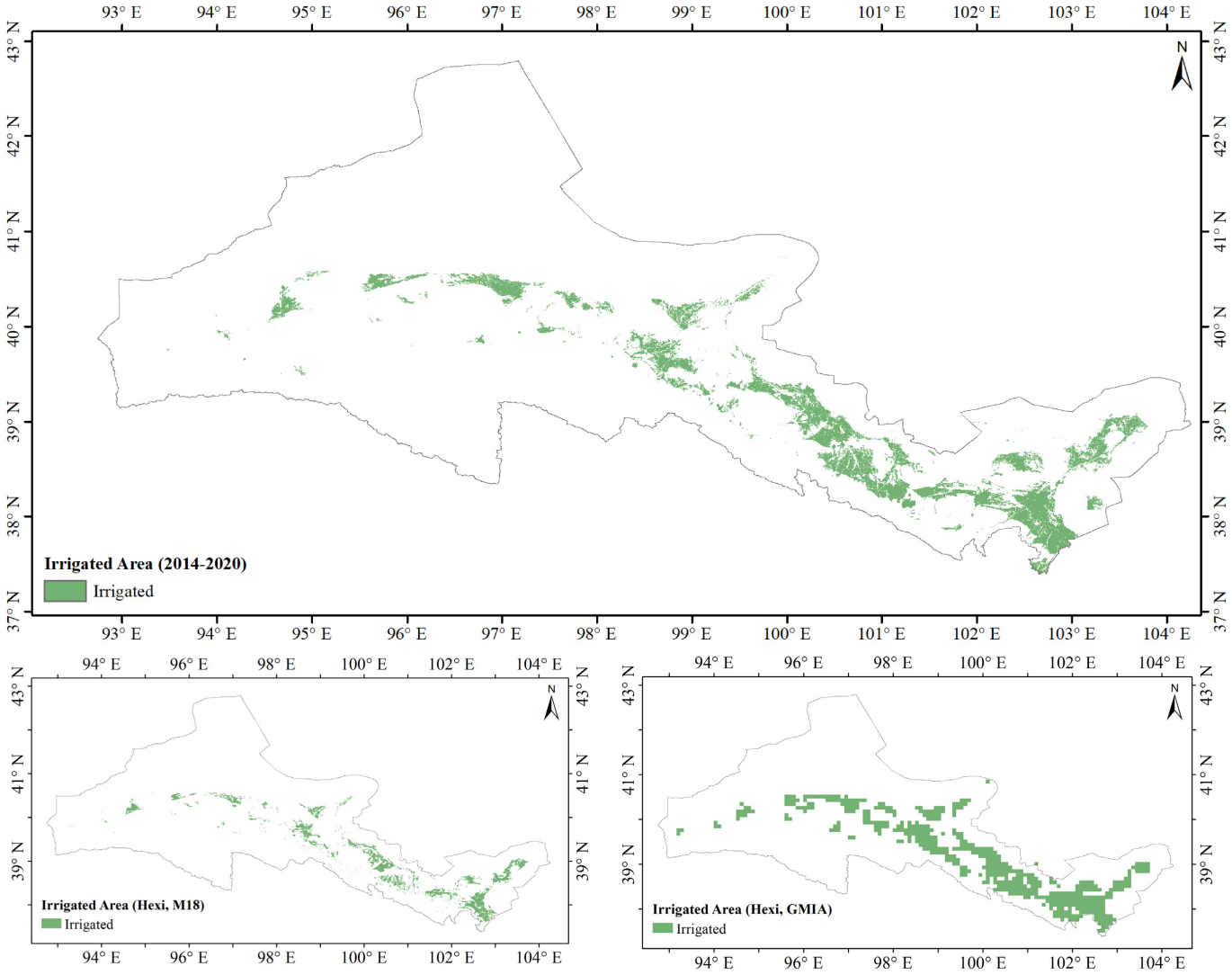


Fig. 5. Irrigated area in the Heihe River Basin mapped by different methods. (Top) Method proposed in this article. (Left Bottom) Method proposed by Meier *et al.* [13]. (Right Bottom) Method proposed by Thenkabail *et al.* [7].

and can be used to map irrigated area with long-term observations. In arid and semiarid regions, agricultural irrigation consumes the most water. The results show that the irrigated cropland in Hexi Corridor was 11 468 km^2 in 1990–1999, 14 463 km^2 in 2012 [13], and 16 308 km^2 in 2014–2020, meaning that the irrigated area increased by 4840 km^2 (42.2%)

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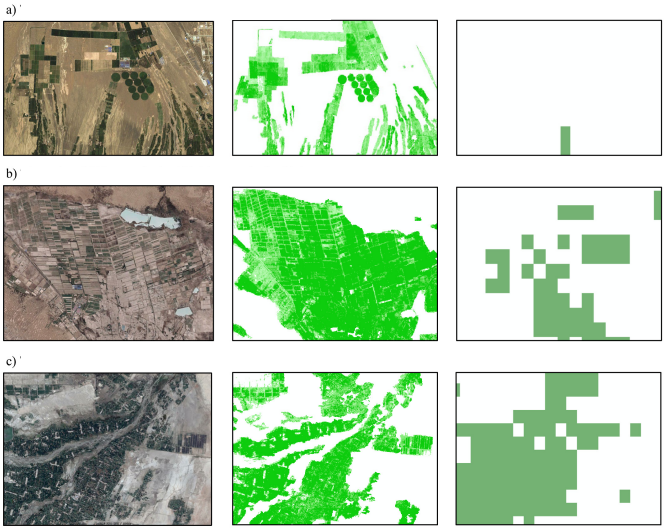


Fig. 6. Spatial characteristics of irrigated area in different regions. Images in each row, from left to right, are: a true-color image via Google Earth, irrigated area mapped by the proposed method, and irrigated area mapped by Meier *et al.* [13]. (a) Time: Sep. 2017, longitude: 100.55°E and latitude: 38.72°N. (b) Time: Apr. 2018, longitude: 100.00°E and latitude: 39.25°N. (c) Time: Jul. 2015, longitude: 98.80°E and latitude: 39.80°N.

over the past 30 years. As shown in Fig. 5, the irrigated area derived with the method proposed in this article is similar to the ones mapped in previous studies [7], [13] but shows more details of the distribution of the irrigated area because of its much higher resolution.

Fig. 6 shows some geometric details of irrigated area in different regions. Irrigation events are not detected in deserts, water bodies, or permanent buildings. In places where center pivot irrigation equipment is used [Fig. 6(a)], a group of circular shapes of irrigation fields is clearly visible. Ridges divide irrigated farmland into regular-shaped blocks [Fig. 6(a) and (b)]. Buildings in villages are distributed in the middle of the farmland [Fig. 6(c)] but cannot be captured by data with medium or coarse resolution.

C. Temporal Change

The left half of Fig. 7 shows the changes in irrigated cropland of Hexi Corridor over the past 30 years. In Hexi Corridor, the area of permanent cropland is 10 269 km^2. Cropland with an area of 1199 km^2 was used for other purposes, and a new area of 6039 km^2 shows the irrigation signal. In general, the irrigated area of Hexi Corridor has rapidly increased in recent years, which is similar to the results concluded in [45] and [46]. This increased water demand for irrigated area poses a challenge for irrigation water supply.

The right half of Fig. 7 represents the changes in three typical regions of Hexi Corridor. Both increase and decrease in the irrigated area have occurred. The pixels with an obviously increasing trend are mainly new irrigated cropland, which are uniformly arranged circles and rectangles shown in Fig. 7(a) and (b). In Fig. 7(c), the number of satellite-based irrigation events has decreased and become urban areas.

V. Discussions

A. Advantages and Limitations of This Study

The most commonly used VI in OPTRAM is NDVI. NDVI has a saturation effect, that is, it cannot reflect its changes well when the vegetation is denser. Some studies have replaced the linear assumption of dry edges and wet edges in OPTRAM with quadratic functions [29]. In this study, the range of NDVI was limited to between 0.3 and 0.7 when calculating parameters for OPTRAM. In this interval, vegetation conditions could be well represented by NDVI. The vegetation condition is similar within 16 days [47]. Thus, the change in soil moisture is only related to the change in STR. In the existing studies, normalized soil moisture is linearly mapped to the range of soil water content at wilting point and soil water content at field capacity [48]. In this study, we use SMAP observations to acquire accurate soil water content at wilting point and field capacity. As shown in Fig. 8, most of the time, the soil water content in cropland does not reach the wilting point or field capacity.

In previous studies, irrigated area is separated from the others using a spectral analysis. There are two disadvantages for this method. On the one hand, similar objects with different spectra and different objects with similar spectrum are common in remote sensing. When the study area is relatively large, the complexity of the underlying surface is significantly increased. Compared with the difference between planting methods (irrigated or rain feed), the difference in crop types has a greater impact on the spectrum. On the other hand, irrigation events are discretely distributed on the time axis. Irrigation events do not occur most of the time, and appropriate images are required for supervised classification. The method proposed in this article overcomes these shortcomings by taking advantage of soil moisture response to irrigation and can be used to map global irrigation areas at field scale.

Hardware issues, such as storage and computing resources, are a huge challenge when processing high-resolution data on a global scale. Although GEE provides much better computing performance than local laptops, the calculation's time complexity and space complexity still need to be considered. In this study, the calculation for parameters of OPTRAM consumes more computational resources than irrigation events retrieval. Thus, the spatial scale effect of OPTRAM parameters needs to be evaluated to balance resource consumption and model accuracy.

R_e is with very low value (14.0%, in general), and it is caused by cloud occlusion and ignored ET. Cloud occlusion has a great influence on optical remote sensing. In five validation sites, about half of ground-truth irrigation events are undetectable for Landsat-8. ET is ignored in (8), which ensures a very high P_e and P_p. We used long-term observations from Landsat to reduce the impact of low recall on irrigated area mapping. However, ET cannot be ignored when estimating the amount of irrigation water. Considering that there is no reliable global ET dataset with high spatiotemporal resolution [44], especially over irrigated areas [49], it is a future work to improve the spatial resolution of ET datasets based on our previous work in developing ET models [49], [50].

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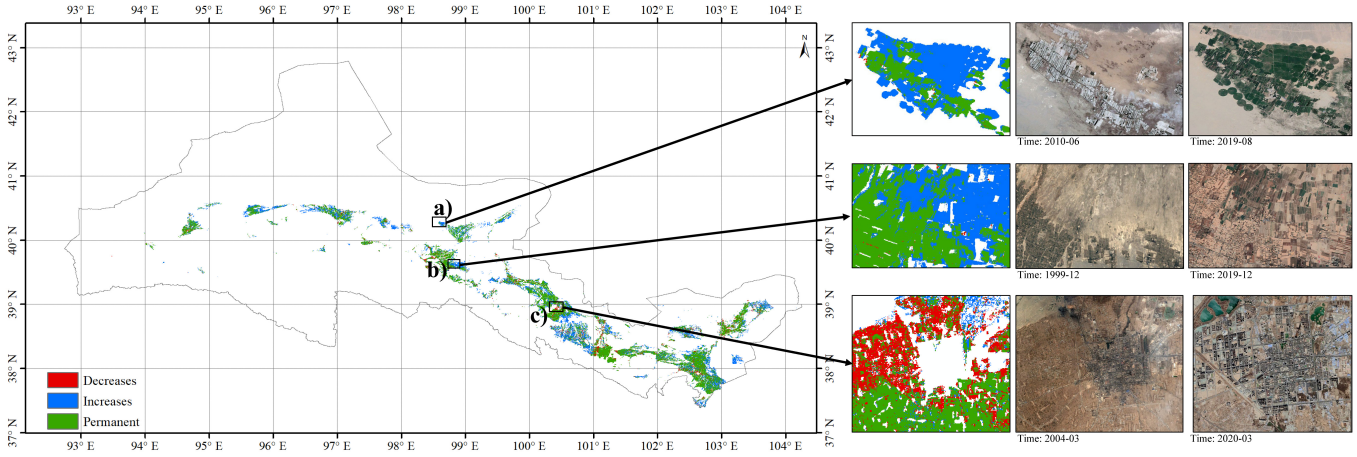


Fig. 7. Difference of irrigated area between two periods: 2014–2020 and 1990–1999 in three regions: (a) Jinta, (b) Suzhou, and (c) Zhangye. Pixels in blue are new irrigated cropland, pixels in red are no longer irrigated, and pixels in green show irrigation signals in both periods. Each line on the right is composed of three images from left to right: the status of irrigation and two true-color images via Google Earth of different periods.

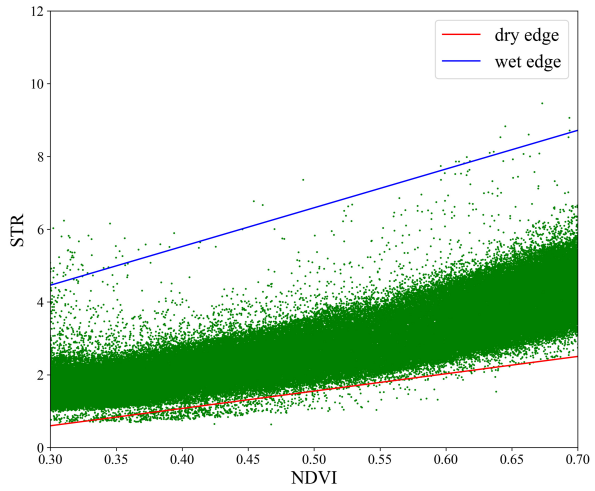


Fig. 8. Scatter of STR-NDVI around Daman superstation.

Although recall is low, the proposed method is still effective in field-scale irrigated area mapping.

B. Time Change

An expansion of urban area, especially in Zhangye, is proven [Fig. 7(c)]. The industrial structure of Hexi Corridor rapidly changes with economic growth and rising water demand [51]. In China, urbanization is one of the reasons for the abandonment of cropland [52]. Water demand in the arid and semiarid areas is mainly used for irrigated agriculture. The area of irrigated areas in Hexi Corridor has increased more than 4000 km^2 in the last 30 years.

A high number of irrigation events are captured in the newly reclaimed cropland. Wasteland could be converted to cropland via agricultural technology, increasing irrigation water demand. In recent decades, irrigation water consumption in arid and semiarid regions has increased faster than humid regions [45]. Water resources management in arid regions especially requires careful consideration as an integrated entity to balance the human–environment interaction.

VI. Conclusion

This study proposes a field-scale irrigated area mapping method, which is based on soil water content change detection using the OPTRAM and requiring only remote sensing data. GEE was used to analyze remote sensing images of Hexi Corridor for the last 31 years. *In situ* validation shows that this method has a precision close to 100% and can be used to map irrigated area with long-time analysis to overcome its low recall in irrigation events. By applying this method to the entire Hexi Corridor for two long periods, we determined that the irrigated cropland area is 16 308 km^2 and has increased by 4840 km^2 (42.2%) over a 31-year time period (1990–2020). The shapes of the irrigated area fit the common characteristics of irrigated cropland, and this irrigated area map in Hexi Corridor provides much more details than coarse-resolution datasets. The increase of irrigated cropland and the expansion of urban area, which is affected by the multiple influences of nature, economy, and policy, have increased the demand for water and pose new challenges to water resources management.

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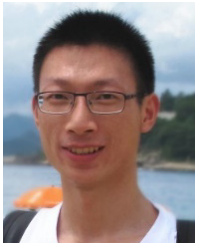
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